

Harish Kumar

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SUMMARY

Backend / Systems Engineer with 7 years of experience building scalable distributed systems. Currently SDE 2 working on large-scale device platforms (50M+ devices). Strong in Java, distributed systems, performance optimization and algorithms.

EXPERIENCE

- Amazon** Chennai, India
Software Development Engineer 2 *Nov 2024 – Present*
 - Member of the Amazon Kindle OS Platform team, contributing to multiple high-impact initiatives across device platforms.
 - Spearheaded Yocto toolchain upgrade from v1.7 to v4 across 50M+ devices, eliminating a critical latency regression caused by OpenSSL v1 to v3 migration through in-depth analysis..
 - Diagnosed and resolved intermittent Kindle sync failure (“last read” issue) potentially impacting all active devices by identifying UDP packet loss in device-server communication and driving a cross-team fix to reduce false offline detection.
 - Delivered multiple features and resolved critical crashes and non-functional issues during 3rd Gen Kindle Scribe bring-up and launch, ensuring production readiness.
 - Designed and implemented chunk-based system update mechanism, optimizing storage utilization and saving up to 350MB per device on 3rd Gen Kindle Scribe.
 - Implemented dark mode support for WebKit-based applications in Kindle device settings for 3rd Gen Kindle Color Scribe.
 - Designed and built device-level battery drain monitoring dashboard with new telemetry metrics, enabling data-driven debugging of power issues across millions of active Kindle devices (Scribe, Paperwhite)
 - Re-architected factory reset mechanism by replacing legacy userspace wipe with Initramfs-based partition wipe, improving system reliability and security.
- Zoho Corporation** Chennai, India
Member of Technical Staff *May 2019 – Nov 2024*
 - Developed and maintained features for Zoho CRM (flagship product), specializing in the Emails module with a focus on scalability and reliability.
 - Revamped Email Sharing (core security feature) by designing REST APIs, migrating from Handlebars.js to Ember.js, and introducing role-based access control, supporting thousands of active users and contributing to a key Zoho CRM 2021 release.
 - Led integration of ZeptoMail (email service platform) into Zoho CRM as DRI, owning cross-service communication, authentication, authorization, and outbound email delivery handling 2M+ emails/day.
 - Owned and enhanced email tracking system processing 1M events/day, enabling real-time insights into email opens, link clicks, and spam reports for customer engagement analytics.
 - Designed and implemented distributed OAuth token generation system using Redis and JVM caching, ensuring race-condition-safe authentication across nodes, supporting 2M+ email API calls/day and 3M+ file upload requests/day.
 - Developed CASL-compliant email footer feature with WYSIWYG editor, enabling customizable and compliant email communication.
 - Owned Mass Mailer system using Kafka for high-throughput batch email delivery, enforcing tenant-level quotas of 250–10K emails/day per account.
 - Led end-to-end revamp of Email and Inventory Templates module, mentoring 2 engineers and driving system design and implementation.
 - Migrated high-traffic (100K+ DAU) WYSIWYG and drag-and-drop template editors from JSP to Ember.js, designed new REST API contracts, and built criteria-based dynamic templating system for flexible email customization.

SKILLS

- Languages:** Java, JavaScript/ TypeScript/ Node.js, C/C++, Python, SQL, Scala, Golang
- Backend/Systems:** Distributed Systems, Kafka, Redis, MySQL, gRPC, Spring Boot, Spring MVC
- Frontend:** React, React Native, Ember.js, HTML/CSS
- Infrastructure:** AWS (S3, Lambda, ECR), Docker
- Systems:** Yocto, Linux, Bitbake
- Methodologies:** Scrum, SDLC
- Tools:** Jira **AI/ LLM Tools:** Agentic AI

EDUCATION

- Rajalakshmi Engineering College** Chennai, India
B.E. Electronics and Communication Engineering *2019*